Federica Domecq Lacroze

domefg26@wfu.edu | (934) 204-8192 | LinkedIn | Website | Madrid, Spain | Winston-Salem, NC

EDUCATION

Wake Forest University — Winston-Salem, NC
M.S. Statistics, Teaching Assistantship — Aug. 2026 – May 2028
Pomona College — Claremont, CA
B.A. Mathematics & Computer Science, Magna Cum Laude; GPA: 3.978/4.00 — Aug. 2022 – May 2026
Honors: Phi Beta Kappa; Sigma Xi; Bruce Jay Levy Senior Prize in Mathematics; Distinction in Mathematics Senior Exercise; Bernard Charnwut Chan Leadership & Service Award
Coursework: Statistical Theory, Computational Statistics, Statistical Learning, Machine Learning, Data Visualisation, Time Series, Linear Models, Probability, Operations Research, Advanced Algorithms, Real Analysis I–II, Linear Algebra

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, GPyTorch, Shapely, matplotlib/seaborn), R (tidyverse, ggplot2), SQL, JavaScript/D3, Haskell, LaTeX
Tools: Git, QGIS, Excel, Linux, HPC/Parallel Computing, LLM-assisted coding & documentation

RESEARCH & EXPERIENCE

Wake Forest University — Winston-Salem, NC

Research Assistant – AI Tools for Experiential Learning in Statistics — Sep. 2026 – Present
- Develop AI-powered tools for practicing statistical consulting skills including client communication, requirements elicitation, team decision-making, and statistical analysis planning; contribute to experimental design, data collection, and analysis.

Graduate Teaching Assistant, Department of Statistical Sciences — Aug. 2026 – Present
- Support instruction for Probability & Statistics and tutor across six statistics and mathematics courses spanning probability, regression, data science, computational statistics, and statistical learning.

Pomona College — Claremont, CA

Kenneth Cooke Research Fellow – "On Par: Leveraging GPR for Perfect Golf, on Average" — May 2025 – Present
- Developed a machine-learning framework for player-specific golf strategy optimization; won **1st Place** in the ASA USRESP Undergraduate Research Project Competition, presented at eCOTS 2026, and preparing a manuscript for peer review.
- Built a Python/QGIS spatial pipeline and Gaussian Process Regression models (`GPyTorch`) to estimate continuous scoring-risk surfaces across 280+ course locations; fused simulated and observed data through a multi-fidelity framework.
- Ran parallel Monte Carlo simulations on HPC infrastructure to optimize club and aim-point decisions under expected-strokes and birdie-probability objectives.

Senior Thesis in Mathematics – Distinction — Jan. 2024 – May 2026
- Developed combinatorial and topological proofs extending Arrow's Impossibility Theorem and analyzed electoral-system and democratic-freedom data across 170+ countries.

Statistics, Data Science & Probability Teaching Assistant — Jan. 2024 – May 2026
- TA across Probability (3×), Introductory Statistics (2×), and Data Science (1×), teaching probability, inference, prediction, data wrangling, and reproducible analysis in `R`/tidyverse.
- Mentored students in R/tidyverse data pipelines, personalised code reviews, and environment setups (RStudio/Positron).

Summer Research Assistant — May 2024 – Aug. 2024
- Modeled gene-expression trajectories in R; developed bootstrap bias corrections and parallelized large-scale simulations across HPC clusters to accelerate model evaluation.

LEADERSHIP & ATHLETICS

Pomona-Pitzer Women's Varsity Golf – *Two-Year Team Captain* — 2022 – 2026
3× NCAA National Championship appearances; 2 individual collegiate wins; 2× Team Conference Champion; 3× All-Conference; 2× All-Region; 4× WGCA All-American Scholar

Association for Women in Mathematics – Co-President 2023 – 2025 **Math Department Liaison** 2025 – 2026

LANGUAGES

English (Fluent) | Spanish (Fluent) | French (Intermediate) | Portuguese (Intermediate)